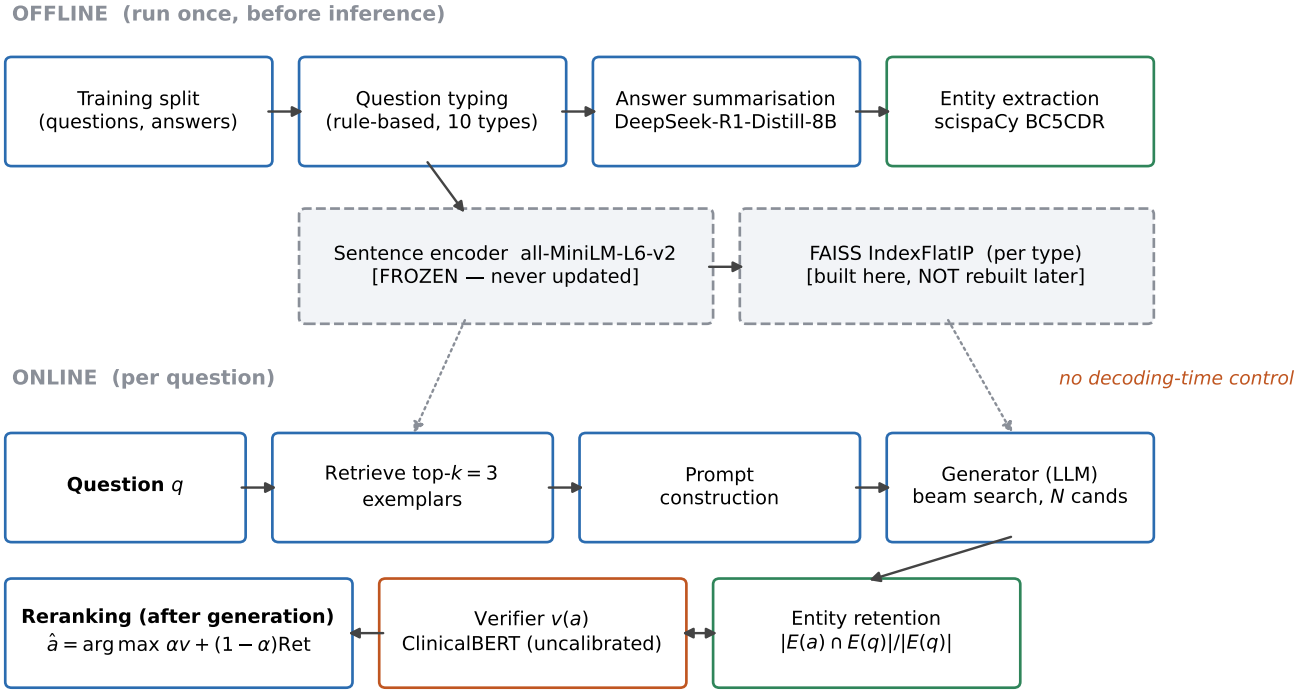
OFFLINE (run once, before inference)
Training split
(questions, answers)
Question typing
(rule-based, 10 types)
Answer summarisation
DeepSeek-R1-Distill-8B
Entity extraction
scispaCy BC5CDR
Sentence encoder all-MiniLM-L6-v2
[FROZEN — never updated]
FAISS IndexFlatIP (per type)
[built here, NOT rebuilt later]
ONLINE (per question)
no decoding-time control
Question q
Retrieve top-$k = 3$
exemplars
Prompt
construction
Generator (LLM)
beam search, N cands
Reranking (after generation)
$\hat{a} = \arg\max \alpha v + (1-\alpha)\text{Ret}$
Verifier $v(a)$
ClinicalBERT (uncalibrated)
Entity retention
$|E(a) \cap E(q)|/|E(q)|$